

Pathmika Weeraratna

Associate Tech Lead | AI/ML Engineer

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Summary

AI/ML engineer and Associate Tech Lead with five years of software engineering experience, having self-initiated and led applied AI/ML capabilities in telecom fraud detection from the ground up at Omobio, progressing from individual learning to leading an AI/ML R&D team. Contributed to 20+ client and product engagements across delivery, architecture, and pre-sales in South Asia, Africa, and the Middle East. Hands-on experience building NLP/embedding-based detection, a GenAI subsystem on local LLMs, and ensemble ML classifiers, backed by strong software engineering fundamentals across full-stack and backend development. Currently pursuing an MSc in Artificial Intelligence at the University of Moratuwa. Recognized as Emerging Employee of the Year (2023).

Experience

Associate Tech Lead

Jan 2026 - Sep 2026

Omobio (Pvt) Ltd

- Led the AI/ML R&D team enhancing an existing rule-based SMS Firewall product with machine-learning-driven detection capabilities, mentoring the cross-functional engineering team of around 10.
- Served as technical subject-matter expert for AI/ML pre-sales engagements, producing architecture proposals and technical validation for telecom fraud-detection and predictive-analytics platforms across Africa, South Asia, and the Middle East.
- Drove solution design and delivery of AI/ML applications through CI/CD-driven workflows.
- Leveraged AI tools throughout the design and development workflow to boost personal productivity, enabling faster turnaround and higher-quality output.
- Also led the SOR (Steering of Roaming) centralized reporting platform, a full-stack telecom data-reporting solution spanning backend pipelines and web dashboards.

Senior Software Engineer

Jan 2024 - Jan 2026

Omobio (Pvt) Ltd

- Self-initiated and led development of a modular SMS Firewall AI/ML Suite (FastAPI, containerized modules, single admin CMS) covering OTP-bypass, phishing-URL, encoded-Unicode, grey-route, and campaign detection; built the OTP-bypass/grey-route module combining SentenceTransformer embeddings (`all-MiniLM-L6-v2`) and cosine-similarity matching against a rolling reference table.
- Engineered the suite's GenAI subsystem: a local-LLM layer (Google Gemma via Ollama, orchestrated with LangChain) generating unallowed-content variants for content review and translation, backed by multilingual BAAI/bge-m3 embeddings for semantic similarity search, and a FastText/NLLB pipeline for language identification and translation.
- Built the suite's phishing-URL detection module: an ensemble ML classifier combining XGBoost and CatBoost with SHAP explainability and Optuna hyperparameter tuning; authored a TPS-based hardware-sizing model with GPU-tier guidance and a tiered data-retention design that substantially reduced steady-state database growth.
- Contributed AI/ML risk-scoring (Random Forest, social-graph SIM-box-ring detection) to an R&D flash-call fraud-detection system on CAMEL/CAP signaling.
- Built a working churn/CLV-prediction prototype (scikit-learn Random Forest) supporting a multi-module predictive-analytics platform proposal (see Key Projects).
- Mentored a team of 10 junior developers on ML engineering practices, code review, and CI/CD (GitHub Actions).

Software Engineer

May 2022 - Jan 2024

Omobio (Pvt) Ltd

- Led design and development of cross-platform mobile applications (Hutch Selfcare, Vodafone Selfcare, MyDialog, RetailHub, Circadian, SFA) in React Native, managing releases across Play Store, App Store, and Huawei AppGallery.
- Built web applications in Next.js and React.js for enterprise and telecom clients, forming the full-stack foundation later applied to AI/ML work.
- Received the Emerging Employee of the Year Award (2023) for outstanding technical contribution.

Software Engineer (Intern) - Full Stack

Oct 2021 - Apr 2022

Omobio (Pvt) Ltd

- Supported development of frontend, backend, and mobile features for enterprise-grade projects using React.js, React Native, and RESTful APIs; transitioned to a full-time Software Engineer role based on strong performance.

Key Projects

- **SMS Firewall AI/ML Suite:** Modular FastAPI-based detection suite (containerized modules, single admin CMS) for OTP-bypass, phishing-URL, encoded-Unicode, grey-route, and campaign detection, processing SMS data streams; phishing-URL detection uses an XGBoost/CatBoost ensemble (SHAP/Optuna), the OTP-bypass/grey-route and encoded-Unicode modules layer in semantic embeddings, and a local-LLM GenAI layer supports suite-wide content review and translation.
- **SOR (Centralized Reporting Platform):** Led a centralized telecom data-reporting platform spanning backend pipelines and web dashboards, with asynchronous task processing and containerized deployment.
- **Flash Call / Voice Firewall ML Layer, 2025:** Random Forest risk scoring and social-graph-based SIM-box-ring detection layered on a CAMEL/CAP signaling fraud-detection system; scoped a next-phase Voice Firewall suite (IRSF, Wangiri, SIM-swap).
- **Predictive CLV & Campaign Insights, 2025:** Scoped a multi-module ML platform architecture (predictive CLV, churn risk, offer recommendation, uplift modeling, behavioral clustering) and built a working Random Forest churn-prediction prototype with a Flask API and React GUI.

Additional Experience & Engagements

Beyond AI/ML, built production mobile and web applications in React Native and Next.js/React, including a custom native alarm engine, HOPP's payment-integrated web portal, and 2+ years owning Hutch Selfcare (RN version upgrade 0.66.5 → 0.76.5, 3+ feature change requests), forming the full-stack fundamentals underpinning the ML work above. Also contributed pre-sales input to several further AI/ML proposals across telecom fraud detection, roaming analytics, and predictive/customer analytics.

Areas of Expertise

- **AI/ML & Data Science:** PyTorch, Hugging Face Transformers, LangChain, Ollama, XGBoost, CatBoost, scikit-learn, SHAP, Optuna, Random Forest
- **NLP & GenAI:** SentenceTransformers (BAAI/bge-m3), semantic search, FastText, NLLB translation, local-LLM deployment (Llama 3, Google Gemma), RAG (Haystack)
- **Applied ML Domains:** Fraud/anomaly detection, predictive modelling (CLV/churn), social-graph analysis, recommendation systems
- **Backend & APIs:** Python, FastAPI, REST API design, Java (Spring Boot), .NET (ASP.NET Core), Microservices
- **Data Engineering:** PostgreSQL, ClickHouse, MariaDB, Elasticsearch/ELK, Apache Airflow, Apache Spark, Celery
- **Web & Mobile:** React.js, Next.js, TypeScript, React Native
- **DevOps:** Microsoft Azure (AZ-900 Certified), Docker, Git, CI/CD (GitHub Actions, Jenkins), Linux

Education

Master of Science in Artificial Intelligence (Reading) - University of Moratuwa July 2025 - Present (Expected 2027)

B.Sc. in Information Systems, Second Class Upper Division, GPA 3.44 - University of Colombo School of Computing 2019 - 2022

Diploma in English (Certificate Level), Distinction Pass - Seagis Campus, Nugegoda 2018

Certifications

Microsoft Certified: Azure Fundamentals (AZ-900) - [Verify Credential](#) Aug 2026

Achievements

- Emerging Employee of the Year - Omobio (2023): recognized for exceptional job performance and contribution to key software engineering projects.
- Director's List Awardee - University of Colombo (2021): achieved GPA above 3.7 in first academic year.